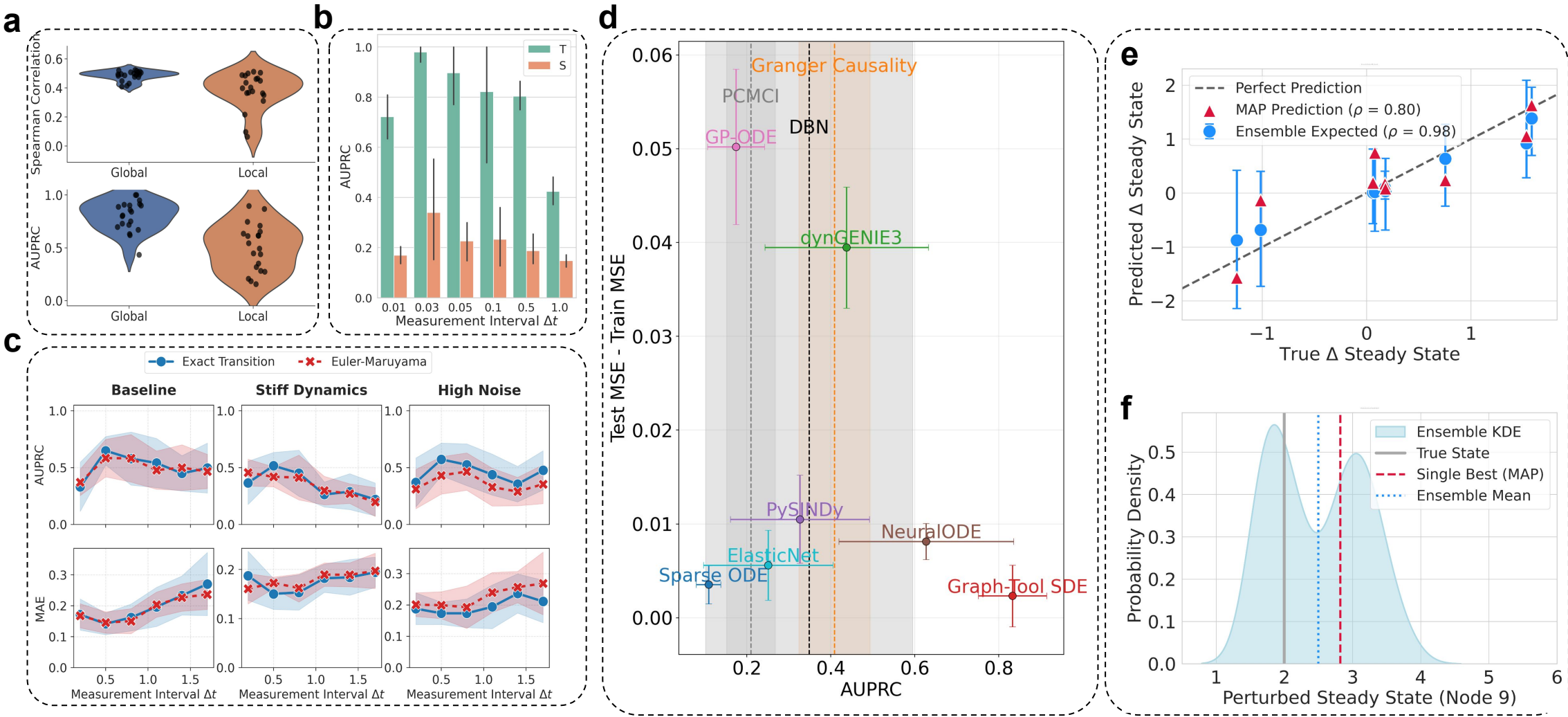

a
Spearman Correlation
Global
Local
AUPRC
Global
Local
b
AUPRC
T
S
0.01 0.03 0.05 0.1 0.5 1.0
Measurement Interval Δt
c
Exact Transition
Euler-Maruyama
Baseline
Stiff Dynamics
High Noise
AUPRC
MAE
Measurement Interval Δt
Measurement Interval Δt
Measurement Interval Δt
d
Granger Causality
PCMCI
DBN
GP-ODE
dynGENIE3
PySINDy
NeuralODE
ElasticNet
Sparse ODE
Graph-Tool SDE
Test MSE - Train MSE
AUPRC
e
Perfect Prediction
MAP Prediction (ρ = 0.80)
Ensemble Expected (ρ = 0.98)
Predicted Δ Steady State
True Δ Steady State
f
Ensemble KDE
True State
Single Best (MAP)
Ensemble Mean
Probability Density
Perturbed Steady State (Node 9)